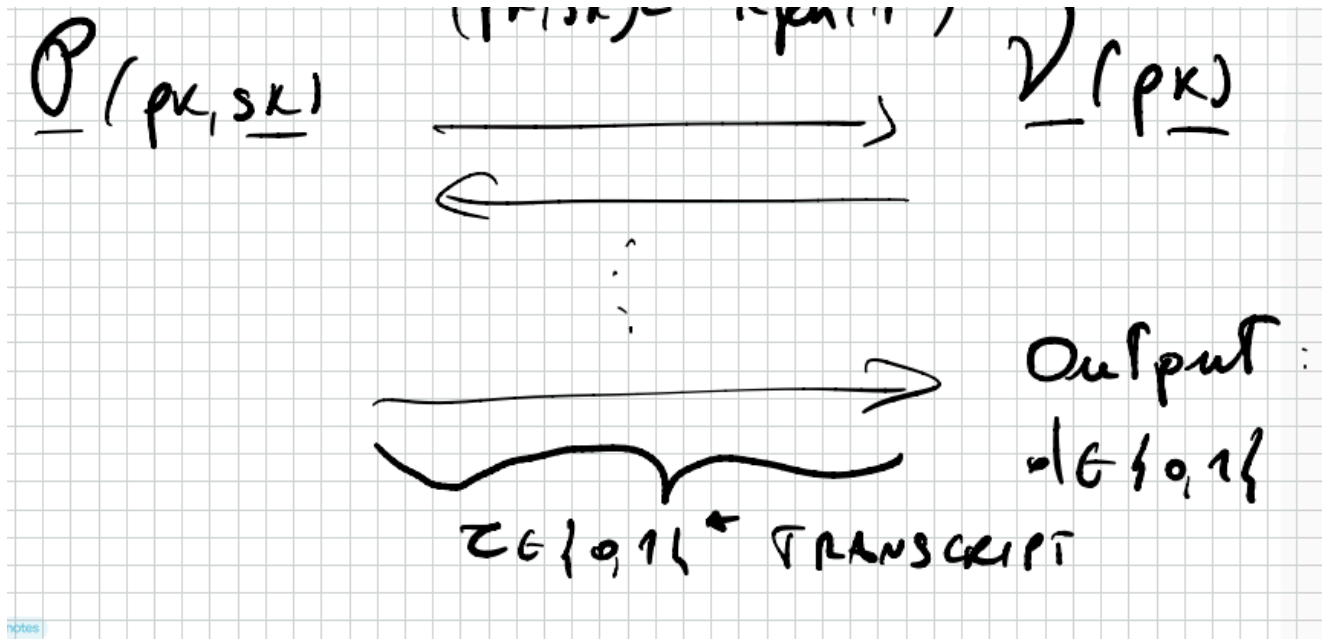


Cryptography - 28_11

Let's start from **Fiat-Shamir Signatures**. This is a standard approach to construct DS from identification scheme.

$$(pk, sk) \leftarrow KGen(1^\lambda)$$



We have two parts:

- Prover ($\mathcal{P}$): wants to show his identity, and has (pk, sk) .
- Verifier ($\mathcal{V}$): wants to verify identity, only knows public key.

The protocol produces a **transcript** (the registration of the messages' exchange), given by:

$$\tau \leftarrow \langle \mathcal{P}(pk, sk) \rightleftharpoons \mathcal{V}(pk) \rangle$$

At the end of the interaction, the Verifier emits $d \in \{0, 1\}$, with 0 being refusal, and 1 acceptance.

Correctness: If $\mathcal{P}$ and $\mathcal{V}$ are honest, the Verifier has to always accept (w.p. 1)

$$\forall (pk, sk) \in KGen(1^\lambda) : Pr[d = 1 | \tau \leftarrow \text{interaction}] = 1$$

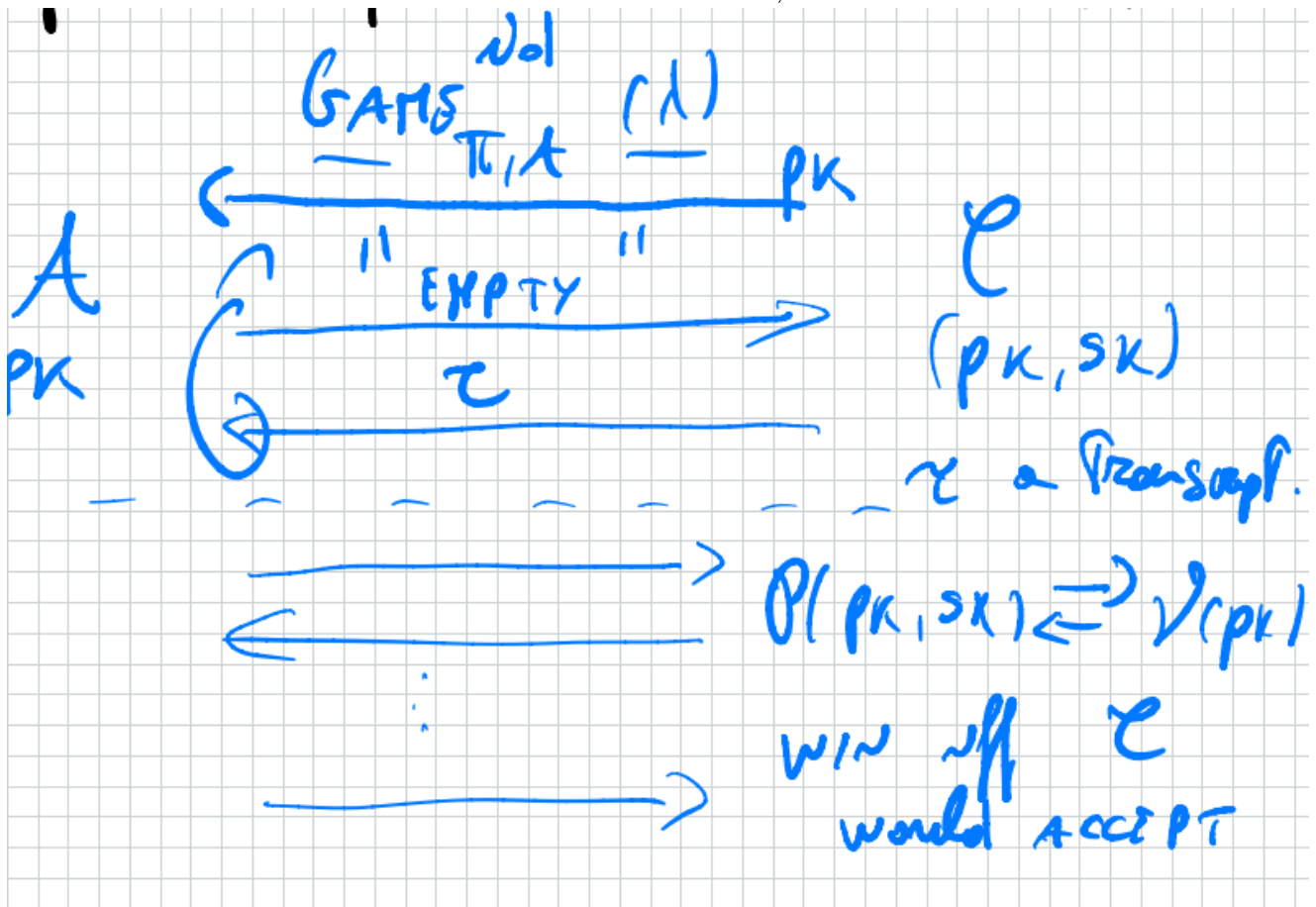
So if the prover is honest, the probability that the verifier outputs a verdict of $d = 1$ is exactly 1, as the prover is honest.

As a passive security: if the attacker does not know the sk , it can't impersonate $\mathcal{P}$. This should be true, even given many accepting transcripts between honest $\mathcal{P}, \mathcal{V}$.

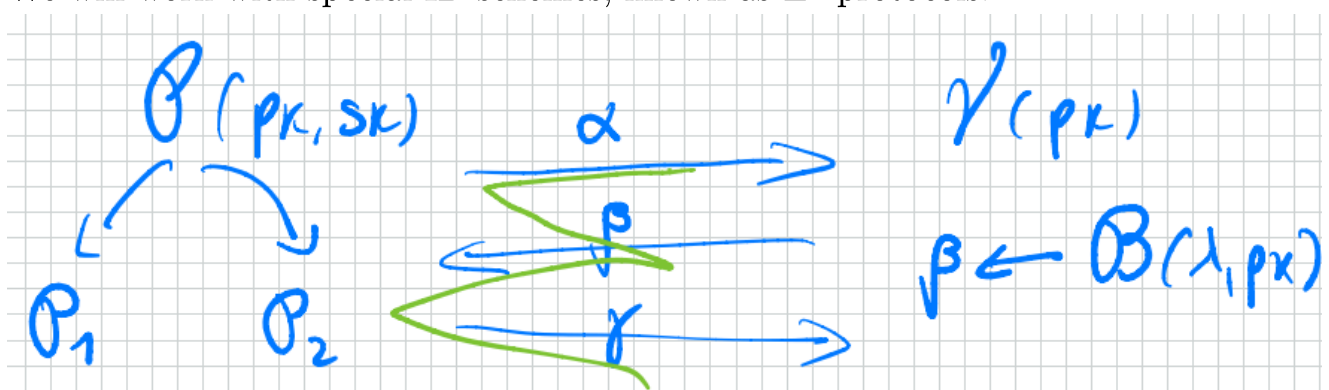
So, even if an attacker $\mathcal{A}$ can observe multiple accepted transcripts, it should not be able to impersonate a Prover.

Here is the GAME.

The probability of the attacker winning $\text{GAME}_{\Pi, \mathcal{A}}^{\text{id}}(\lambda)$ should be negligible.



We will work with special ID schemes, known as Σ -protocols.



These are three movement protocols:

1. Commitment (α): the prover $\mathcal{P}$ sends a message α (known as commitment).
2. Challenge (β): the verifier $\mathcal{V}$ selects a random value β (the challenge), sending it to $\mathcal{P}$.
3. Response (γ): The prover computes an answer γ using his secret key sk , α and β .

To conclude, the verifier checks if (α, β, γ) is valid.

We have a non-triviality property:

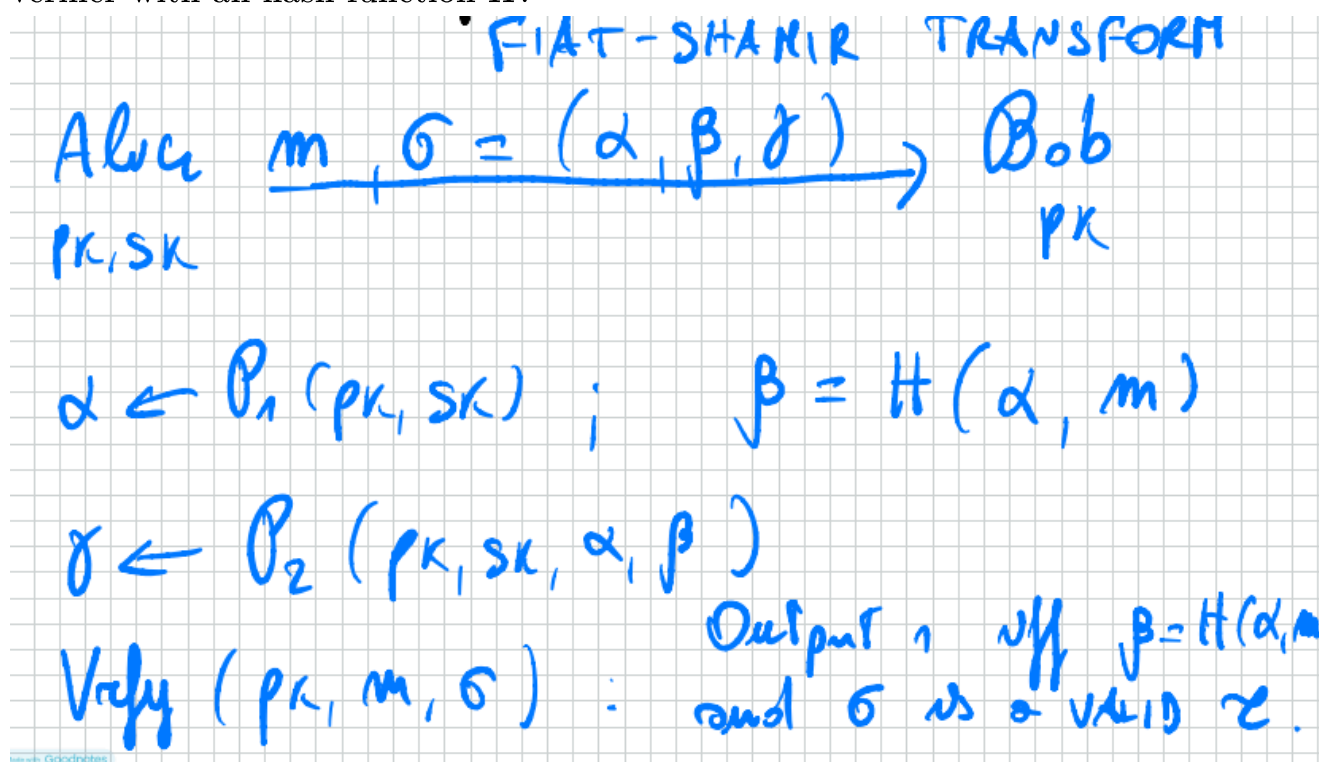
Given $\alpha \leftarrow \mathcal{P}_1(pk, sk)$:

$Pr[\alpha = \hat{\alpha}] \leq \text{negl}(\lambda)$, this $\forall pk, sk, \hat{\alpha}$.

It should not be easy to find a valid answer without knowing the secret.

Now, given a Σ -protocol, we can easily obtain a Digital Signature in the Random Oracle Model.

To do so, we use the **Fiat-Shamir transform**. The idea is to replace the human verifier with an hash function H .



How does it work?

Let's say that Alice wants to sign a message m using (pk, sk) . Alice executes the first part of the ID protocol, computing α . Rather than waiting β from the verifier, Alice computes the challenge by herself.

$$\beta = H(\alpha, m)$$

She then computes γ exactly as in the ID protocol.

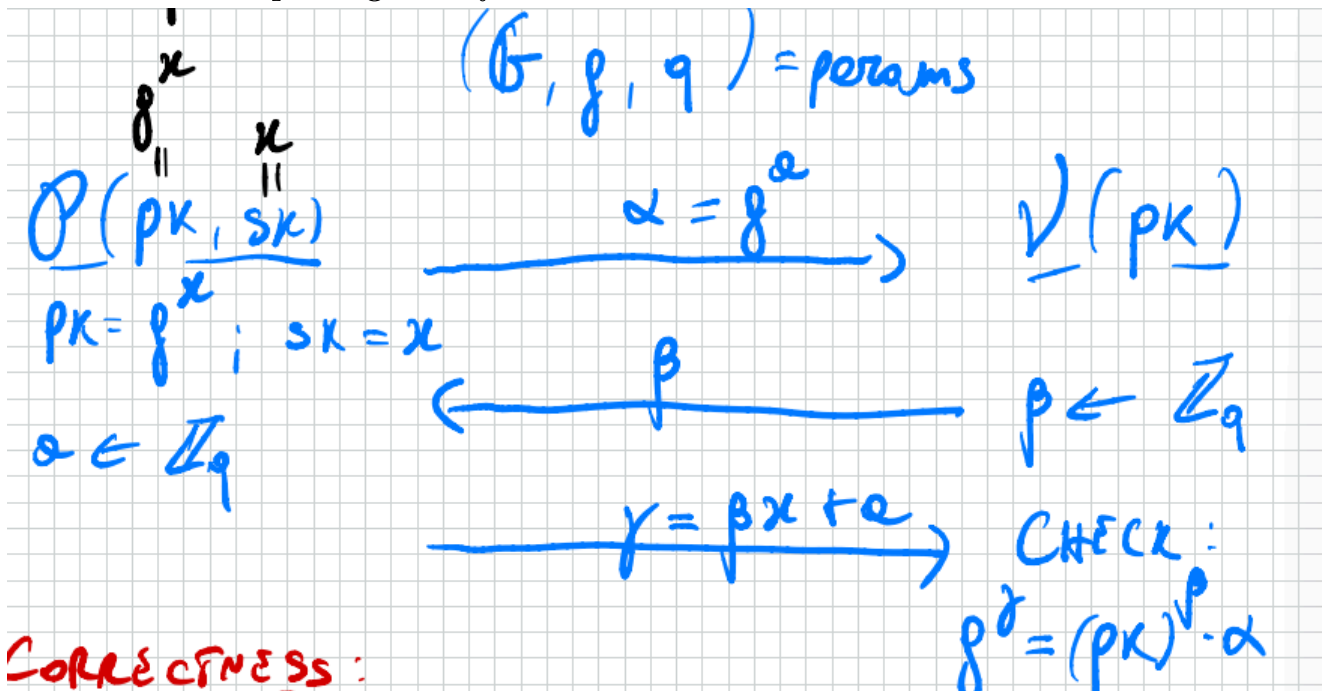
The digital signature σ is the couple (or triplet) that allows verification.

Usually:

$$\sigma = (\alpha, \beta, \gamma)$$

To verify: Bob receives m and σ . He computes $\beta' = H(\alpha, m)$, and verify if (α, β', γ) is a valid transcript for the ID scheme.

A concrete example is given by **Schnorr Σ -Protocol**.



We take a cyclic group $\mathbb{G}$ with order q , and generator g .

We choose as keys $sk = x$ (with $x \leftarrow \mathbb{Z}_q$), $pk \leftarrow g^x$.

The protocol then works in this way:

1. Commitment ($\mathcal{P} \rightarrow \mathcal{V}$): the prover chooses a random value $a \in \mathbb{Z}_q$, and sends $\alpha = g^a$.
2. Challenge ($\mathcal{V} \rightarrow \mathcal{P}$): the verifier chooses a random value $\beta \in \mathbb{Z}_q$, sending it to $\mathcal{P}$.
3. Response ($\mathcal{P} \rightarrow \mathcal{V}$): the prover computes:

$$\gamma = \alpha + \beta \cdot x \pmod{q}$$

The verifier checks if:

$$g^\gamma \stackrel{?}{=} (pk)^\beta \cdot \alpha$$

Here is the correctness:

$$g^\gamma = g^{\beta \cdot x + \alpha} = (g^x)^\beta \cdot g^\alpha = (pk)^\beta \cdot \alpha$$

✓ Theorem

If Π is passively secure, then Fiat-Shamir signatures are UFCMA in the ROM.

(Proof next time).

Let's focus first on constructing passively secure ID schemes. Two sufficient properties:

- Honest-Verifier zero knowledge
- Special soundness.

Let's define then:

✓ Definition (HVZK)

An ID scheme $\Pi(KGen, \mathcal{P}, \mathcal{V})$ is HVZK if $\exists$ PPT Simulator (Sim) such that:

$$\underbrace{(pk, sk, Sim(pk))}_{\text{simulated } \tau} \approx_c \underbrace{(pk, sk, \mathcal{P}(pk, sk) \rightleftharpoons V(pk))}_{\text{honest } \tau}$$

So a scheme is HVZK if there exists a simulator that, by only knowing the public key, can generate a transcript (α, β, γ) that is indistinguishable from a transcript computed by a real interaction between honest Prover and Verifier.

An example is given by Schnorr:

We can simulate a transcript without knowing x , in fact the Simulator:

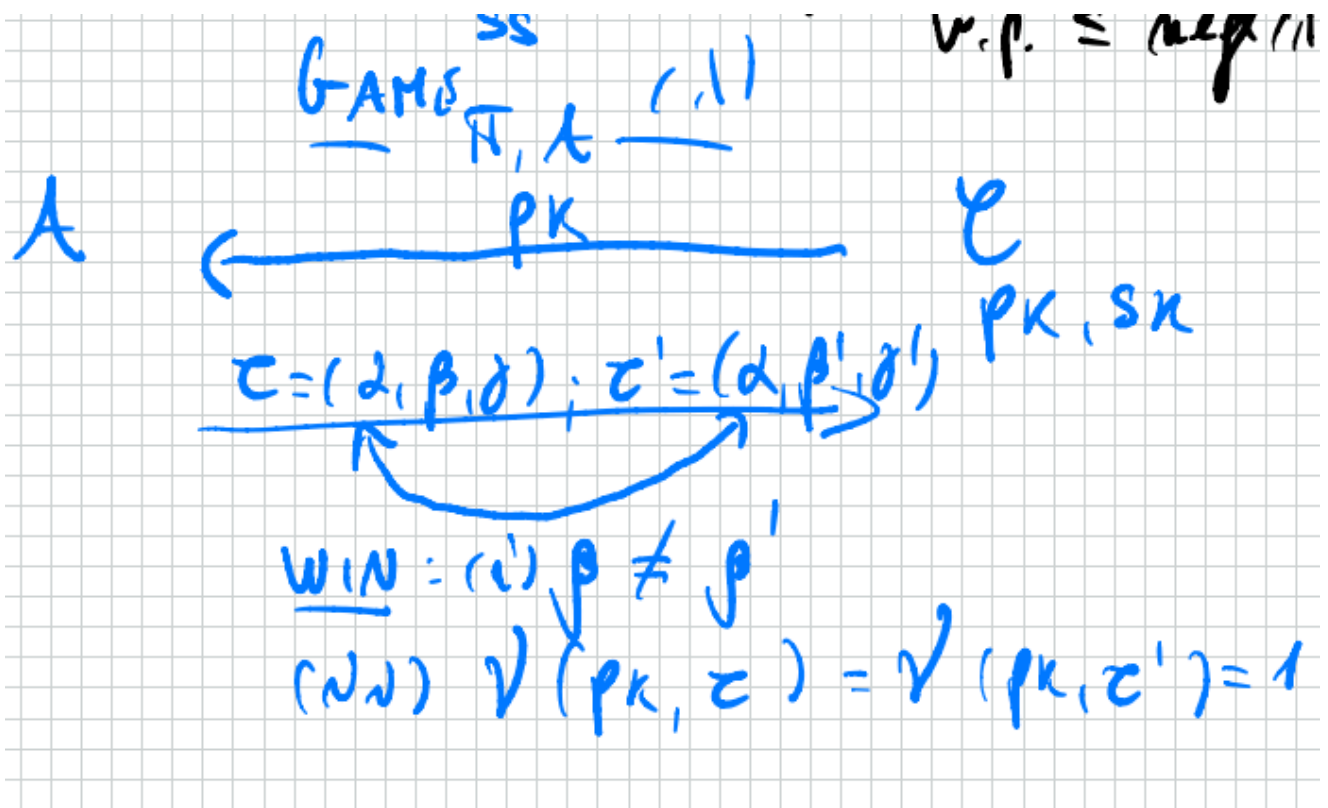
1. Chooses random $\beta \in \mathbb{Z}_q$
2. Chooses random answer $\gamma \in \mathbb{Z}_q$
3. Computes α in a way the equation is true:

$$\alpha = g^\gamma \cdot (pk)^{-\beta}$$

The output is given by (α, β, γ) . The distribution of probability of this simulated transcript is identical to the real one, so Schnorr is HVZK.

✓ Definition (SS)

$\Pi = (KGen, \mathcal{P}, \mathcal{V})$ is special sound if $\forall$ PPT $\mathcal{A}$ the following game can be won w.p. $\leq \text{negl}(\lambda)$.



So a scheme has Special Soundness if, given two valid transcripts τ and τ' , which have the same commitment α but two different challenges $\beta \neq \beta'$, it's possible to extract the secret key sk . This means that a prover cannot answer to two different questions having the same α without knowing the secret key.

Example: Schnorr has this property assuming DL is hard. In fact, given τ, τ' as above we have:

$$\begin{aligned} g^\gamma &= (pk)^\beta \cdot \alpha \\ g^{\gamma'} &= (pk)^{\beta'} \cdot \alpha \\ \implies g^{\gamma - \gamma'} &= (g^x)^{\beta - \beta'} \\ \implies g^x &= g^{(\gamma - \gamma')(\beta - \beta')^{-1}} \end{aligned}$$

And so $x = (\gamma - \gamma')(\beta - \beta')^{-1} \pmod{q}$ since $\beta \neq \beta'$.

✓ Theorem

Special Soundness + HVZK $\implies$ Passive security.